Lab3

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set()
```

Output:

```
Matplotlib is building the font cache; this may take a moment.
```

```
df = pd.read_csv("recent_movies_2024_2026.csv")
df.head()
```

Output:

```
              title  release_year release_month  release_day  \
0  The Mummy Murders          2024       January          2.0
1           DarkGame          2024       January          4.0
2   He Went That Way          2024       January          NaN
3        The Painter          2024       January          NaN
4     The Bricklayer          2024       January          NaN

                     genres  rating_out_of_10  vote_count  runtime_minutes  \
0  Horror, Mystery, Thriller               7.8      280952              138
1  Horror, Mystery, Thriller               7.6      109812              103
2          Sci-Fi, Adventure               7.4      234187              128
3          Sci-Fi, Adventure               9.2      165532              158
4          Sci-Fi, Adventure               7.1      227023               92

   box_office_usd  budget_usd          director              screenwriter  \
0        45264640    12501632               NaN              Will Donahue
1        40487340    11440392    Howard J. Ford  Gary Grant,Niall Johnson
2        29375529    22583686   Jeffrey Darling            Evan M. Wiener
3        28225644    21817096  Kimani Ray Smith          Brian Buccellato
4        29255541    22503694      Renny Harlin   Hanna Weg, Matt Johnson

                                          cast  \
0  Colin Bressler, Leila Annastasia Scott, Jason ...
1  Ed Westwick, Andrew P. Stephen, Natalya Tsvetk...
2      Jacob Elordi,Zachary Quinto,Patrick J. Adams
3  Charlie Weber,Jon Voight,Marie Avgeropoulos, M...
4  Aaron Eckhart,Nina Dobrev,Tim Blake Nelson,Ilf...

                               production_company content_rating language  \
0                             Gravitas Ventures              R  English
1                             Gravitas Ventures              R  English
2  Vertical Entertainment/Mister Smith Entertainment         PG-13  English
3                             Republic Pictures          PG-13  English
4           Vertical Entertainment/Millennium Media          PG-13  English

         country                                          synopsis  \
0  United States  'The Mummy Murders' (2024) is a feature film d...
1  United States  'DarkGame' (2024) is a feature film directed b...
2  United States  'He Went That Way' (2024) is a feature film di...
3  United States  'The Painter' (2024) is a feature film directe...
4  United States  'The Bricklayer' (2024) is a feature film dire...

                                  wikipedia_url  \
0   https://en.wikipedia.org/wiki/The_Mummy_Murders
1            https://en.wikipedia.org/wiki/DarkGame
2    https://en.wikipedia.org/wiki/He_Went_That_Way
3  https://en.wikipedia.org/wiki/The_Painter_(202...
```

```
4       https://en.wikipedia.org/wiki/The_Bricklayer

                                        poster_url
0  https://images.unsplash.com/photo-148959984992...
1  https://images.unsplash.com/photo-148959984992...
2  https://images.unsplash.com/photo-148959984992...
3  https://images.unsplash.com/photo-148959984992...
4  https://images.unsplash.com/photo-148959984992...
```

```
df.dtypes
```

Output:

```
title                  object
release_year            int64
release_month          object
release_day           float64
genres                 object
rating_out_of_10      float64
vote_count              int64
runtime_minutes         int64
box_office_usd          int64
budget_usd              int64
director               object
screenwriter           object
cast                   object
production_company     object
content_rating         object
language               object
country                object
synopsis               object
wikipedia_url          object
poster_url             object
dtype: object
```

```
print("Missing values per column:\n")
print(df.isna().sum())
```

Output:

```
Missing values per column:

title                   0
release_year            0
release_month           0
release_day           676
genres                  0
rating_out_of_10        0
vote_count              0
runtime_minutes         0
box_office_usd          0
budget_usd              0
director              305
screenwriter           22
cast                    0
production_company      0
content_rating          0
language                0
country                 0
synopsis                0
wikipedia_url           0
poster_url              0
dtype: int64
```

```
print("Check Duplicates:\n")
df.duplicated()[df.duplicated()==True]
```

Output:

```
Check Duplicates:
```

Output:

```
Series([], dtype: bool)

print(" Number of rows and columns:\n")

print("Shape (rows, columns): ", df.shape,"\n")

print("number of rows: ", df.shape[0])
print("number of columns: ", df.shape[1])
```

Output:

```
 Number of rows and columns:

Shape (rows, columns):  (916, 20)

number of rows:  916
number of columns:  20

print("Summary statistics for numerical columns:\n")
df.describe(include='all')
```

Output:

```
Summary statistics for numerical columns:
```

Output:

```
             title  release_year release_month  release_day  \
count          916    916.000000           916   240.000000
unique         914           NaN            12          NaN
top     Don't Move           NaN        August          NaN
freq             2           NaN            87          NaN
mean           NaN   2024.975983           NaN    15.679167
std            NaN      0.816143           NaN     9.011452
min            NaN   2024.000000           NaN     1.000000
25%            NaN   2024.000000           NaN     7.000000
50%            NaN   2025.000000           NaN    16.000000
75%            NaN   2026.000000           NaN    24.000000
max            NaN   2026.000000           NaN    31.000000

                  genres  rating_out_of_10     vote_count  runtime_minutes  \
count                916        916.000000     916.000000       916.000000
unique                20               NaN            NaN              NaN
top     Sci-Fi, Adventure               NaN            NaN              NaN
freq                 323               NaN            NaN              NaN
mean                 NaN          7.588755  190883.281659       128.033843
std                  NaN          0.959322   97582.137783        21.704472
min                  NaN          6.000000    7128.000000        90.000000
25%                  NaN          6.800000  119065.500000       110.750000
50%                  NaN          7.600000  179263.000000       129.000000
75%                  NaN          8.400000  253092.000000       146.000000
max                  NaN          9.200000  452490.000000       164.000000

        box_office_usd    budget_usd      director screenwriter  \
count     9.160000e+02  9.160000e+02           611          894
unique             NaN           NaN           551          858
top                NaN           NaN  Renny Harlin  David Koepp
freq               NaN           NaN             5            5
mean      4.112292e+07  2.450549e+07           NaN          NaN
```

Public - عام

```
std       6.774094e+07  2.138200e+07           NaN          NaN
min       2.531260e+07  1.084931e+07           NaN          NaN
25%       2.784818e+07  2.111606e+07           NaN          NaN
50%       2.908222e+07  2.199819e+07           NaN          NaN
75%       3.139283e+07  2.302142e+07           NaN          NaN
max       5.107004e+08  1.712464e+08           NaN          NaN

                                                     cast production_company  \
count                                                 916                916
unique                                                915                663
top     Matthew Rhys,Katee Sackhoff,Titus Welliver,Tro...           Vertical
freq                                                    2                 25
mean                                                  NaN                NaN
std                                                   NaN                NaN
min                                                   NaN                NaN
25%                                                   NaN                NaN
50%                                                   NaN                NaN
75%                                                   NaN                NaN
max                                                   NaN                NaN

       content_rating language        country  \
count             916      916            916
unique              3        1              1
top             PG-13  English  United States
freq              806      916            916
mean              NaN      NaN            NaN
std               NaN      NaN            NaN
min               NaN      NaN            NaN
25%               NaN      NaN            NaN
50%               NaN      NaN            NaN
75%               NaN      NaN            NaN
max               NaN      NaN            NaN

                                              synopsis  \
count                                              916
unique                                             916
top     'The Mummy Murders' (2024) is a feature film d...
freq                                                 1
mean                                               NaN
std                                                NaN
min                                                NaN
25%                                                NaN
50%                                                NaN
75%                                                NaN
max                                                NaN

                                         wikipedia_url  \
count                                              916
unique                                             916
top     https://en.wikipedia.org/wiki/The_Mummy_Murders
freq                                                 1
mean                                               NaN
std                                                NaN
min                                                NaN
25%                                                NaN
50%                                                NaN
75%                                                NaN
max                                                NaN

                                           poster_url
count                                             916
unique                                              1
top     https://images.unsplash.com/photo-148959984992...
freq                                              916
mean                                              NaN
std                                               NaN
min                                               NaN
25%                                               NaN
```

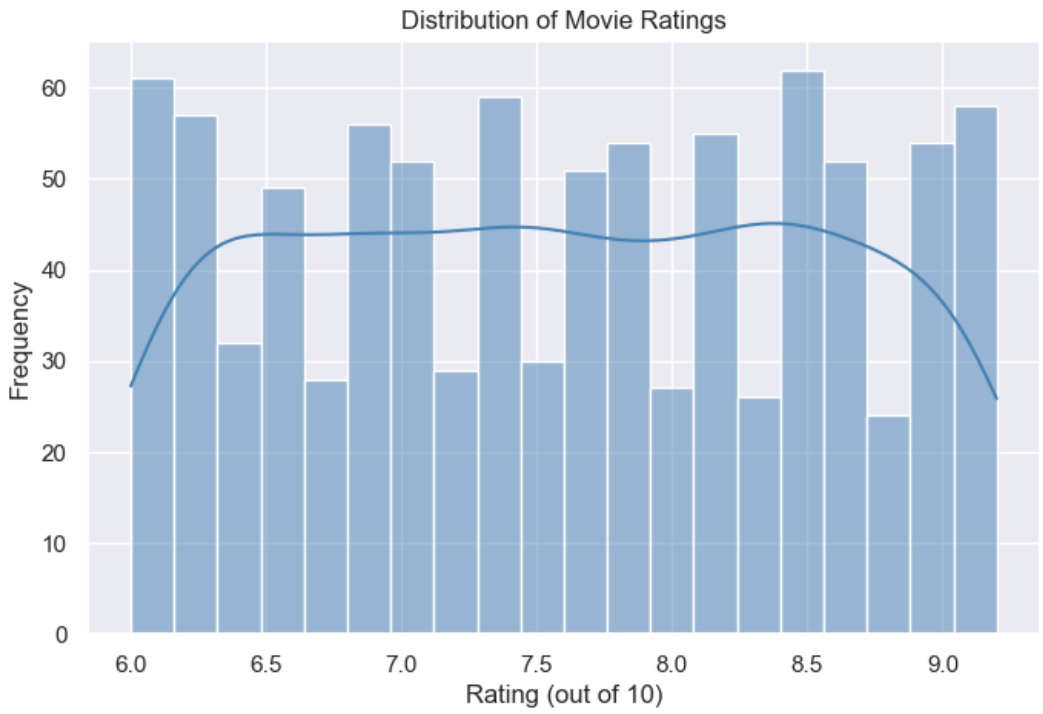
Public - عام

```
50%                                                   NaN
75%                                                   NaN
max                                                   NaN
```

```
plt.figure(figsize=(8, 5))
sns.histplot(df['rating_out_of_10'], bins=20, kde=True, color='steelblue')
plt.title('Distribution of Movie Ratings')
plt.xlabel('Rating (out of 10)')
plt.ylabel('Frequency')
plt.show()
```

Output:

```
<Figure size 800x500 with 1 Axes>
```

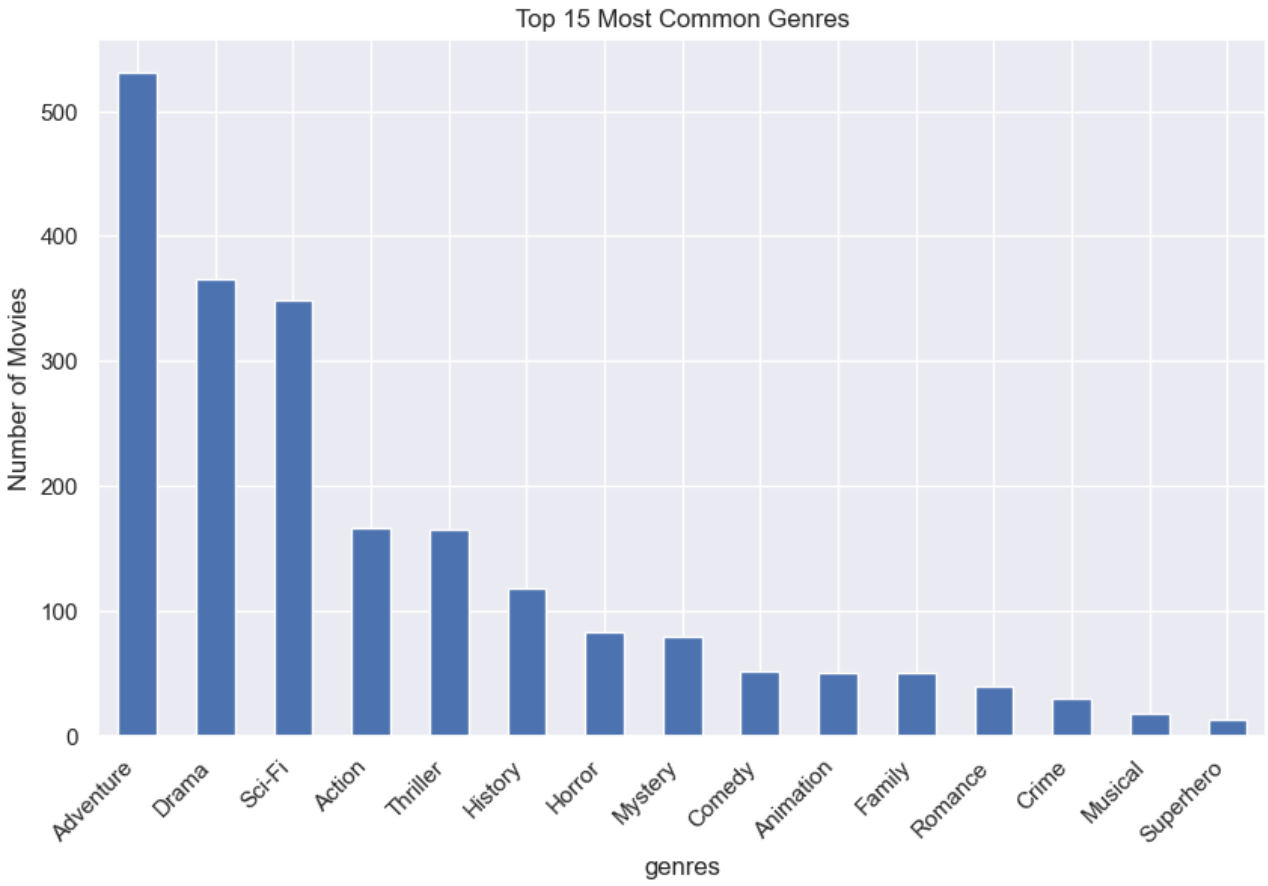



```
from collections import Counter
all_genres = df['genres'].dropna().str.split(',').explode().str.strip()
genre_counts = all_genres.value_counts().head(15)

plt.figure(figsize=(10, 6))
genre_counts.plot(kind='bar')
plt.title("Top 15 Most Common Genres")
plt.ylabel("Number of Movies")
plt.xticks(rotation=45, ha='right')
plt.show()
```

Output:

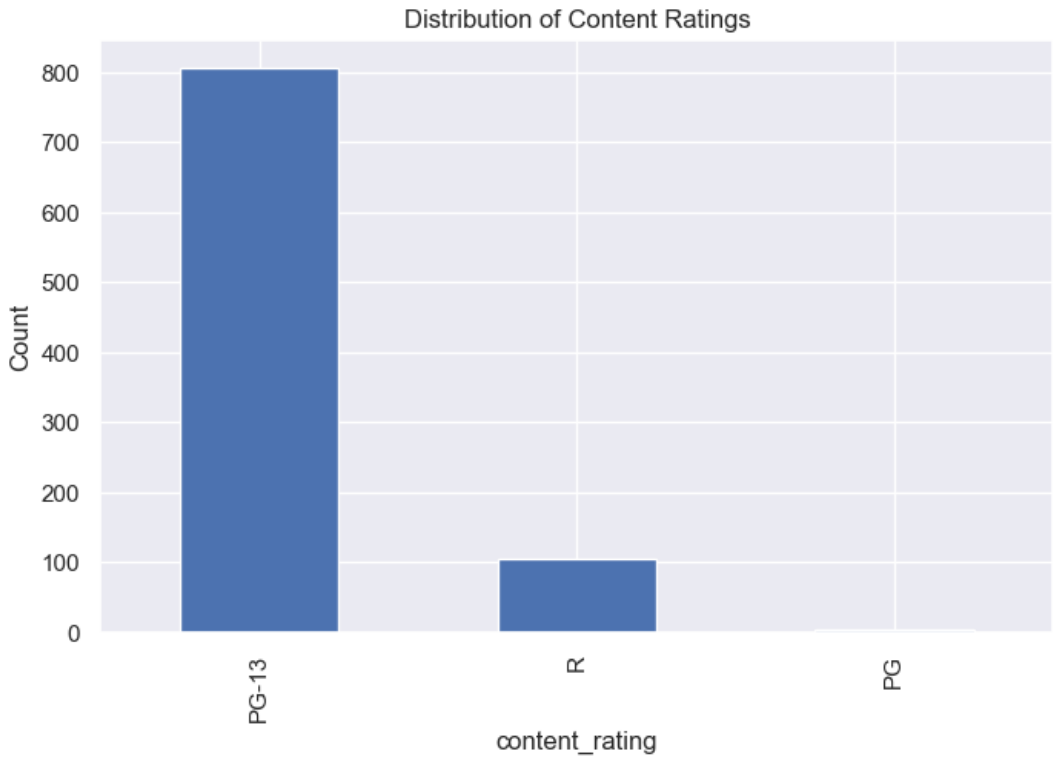
```
<Figure size 1000x600 with 1 Axes>
```




```
plt.figure(figsize=(8, 5))
df['content_rating'].value_counts().plot(kind='bar')
plt.title("Distribution of Content Ratings")
plt.ylabel("Count")
plt.show()
```

Output:

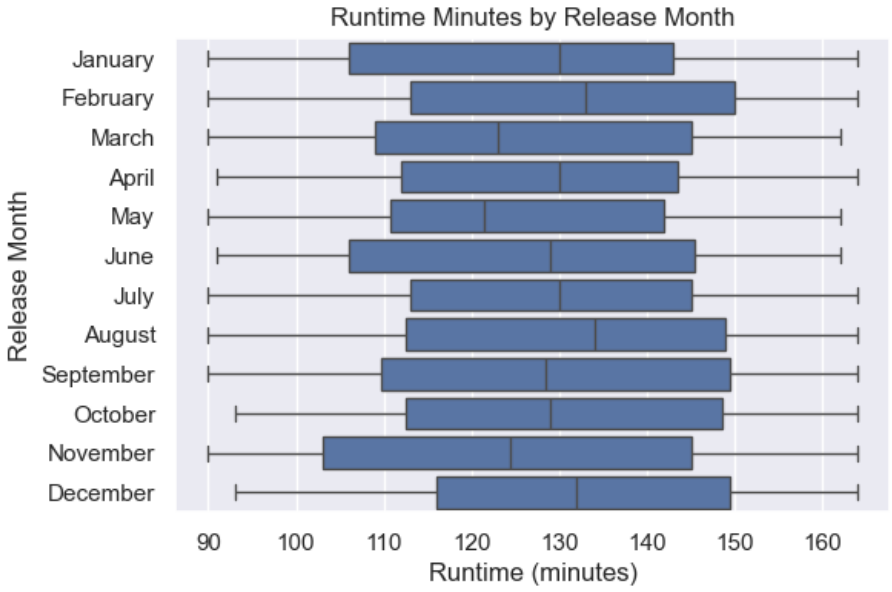
```
<Figure size 800x500 with 1 Axes>
```




```
plt.figure(figsize=(6,4))
sns.boxplot(x='runtime_minutes', y='release_month', data=df)
plt.title('Runtime Minutes by Release Month')
plt.xlabel('Runtime (minutes)')
plt.ylabel('Release Month')
plt.show()
```

Output:

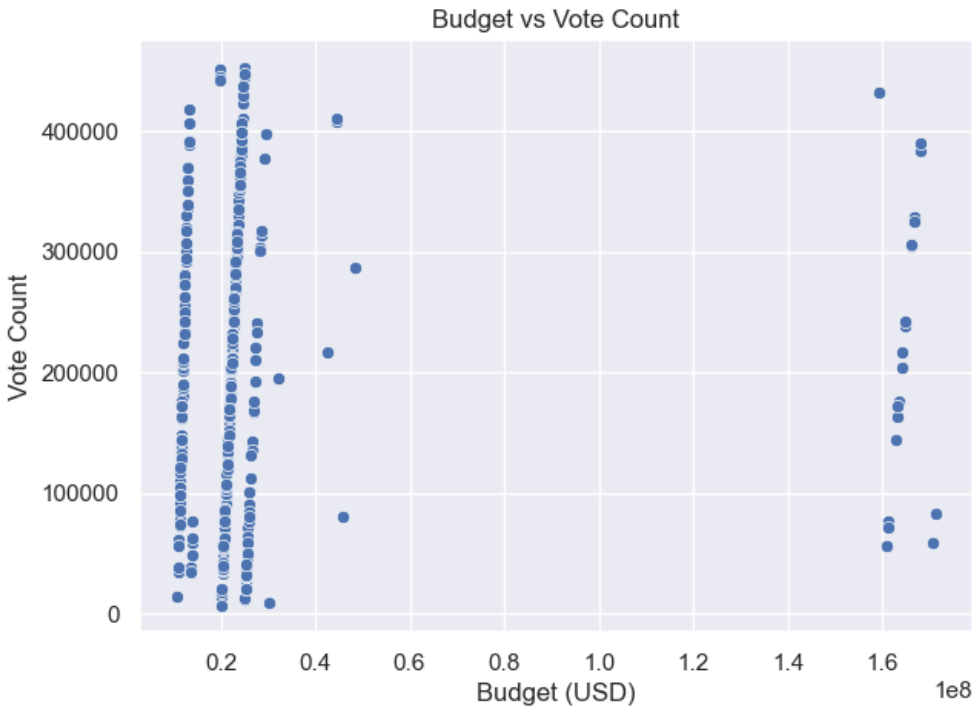
```
<Figure size 600x400 with 1 Axes>
```



```
plt.figure(figsize=(7,5))
sns.scatterplot(x='budget_usd', y='vote_count', data=df)
plt.title('Budget vs Vote Count')
plt.xlabel('Budget (USD)')
plt.ylabel('Vote Count')
plt.show()
```

Output:

```
<Figure size 700x500 with 1 Axes>
```



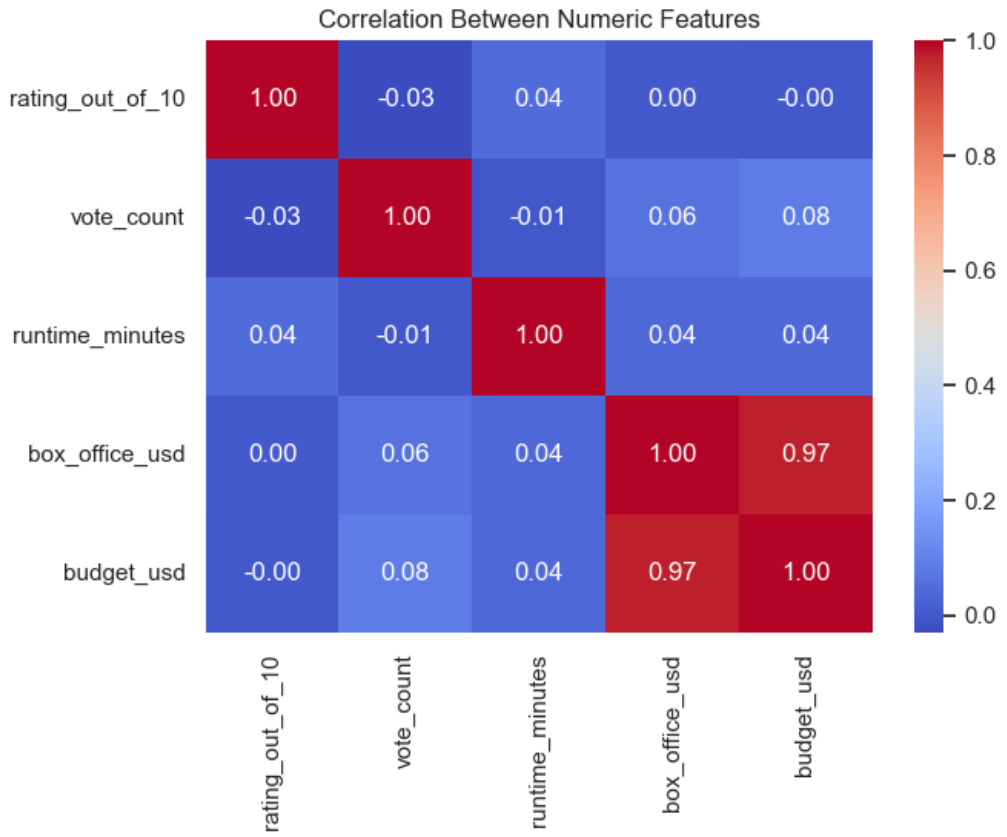

```
numeric_cols = ['rating_out_of_10', 'vote_count', 'runtime_minutes', 'box_office_usd',
'budget_usd']
corr_matrix = df[numeric_cols].corr()

plt.figure(figsize=(7, 5))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Between Numeric Features')
plt.show()
```

Output:

```
<Figure size 700x500 with 2 Axes>
```

Public - عام



Public - عام